

ONLINE MASTERS IN **DATA SCIENCE**

DSC258: NATURAL LANGUAGE PROCESSING

POS TAGGING USING SPACY

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- Part-of-Speech Tags
- **POS Tagging using spaCy**
- Classic POS Tagger using HMM
- More POS Tagging Ideas & Summary

```
In [1]: import spacy
nlp = spacy.load("en_core_web_sm")
```

Part-of-speech tags

```
In [2]: doc = nlp("Autonomous cars shift insurance liability toward manufacturers")
print([token.text for token in doc])
print([token.pos_ for token in doc])
# ['DET', 'VERB', 'DET', 'NOUN', 'PUNCT']
# Fine-grained part-of-speech tags
print([token.tag_ for token in doc])
```

```
['Autonomous', 'cars', 'shift', 'insurance', 'liability', 'toward', 'manu  
facturers']
```

```
['ADJ', 'NOUN', 'VERB', 'NOUN', 'NOUN', 'ADP', 'NOUN']
```

```
['JJ', 'NNS', 'VBP', 'NN', 'NN', 'IN', 'NNS']
```

POS TAGGING USING SPACY

```
In [3]: doc = nlp("Neural Network model predictions get better and better")
```

```
print([token.text for token in doc])
print([token.pos_ for token in doc])
# ['DET', 'VERB', 'DET', 'NOUN', 'PUNCT']
# Fine-grained part-of-speech tags
print([token.tag_ for token in doc])
```

```
['Neural', 'Network', 'model', 'predictions', 'get', 'better', 'and', 'better']
['PROPN', 'PROPN', 'NOUN', 'NOUN', 'VERB', 'ADJ', 'CCONJ', 'ADV']
['NNP', 'NNP', 'NN', 'NNS', 'VBP', 'JJR', 'CC', 'RBR']
```

```
In [4]: doc = nlp("the pilot's approach was off-center. He still managed a three-point landing")
```

```
print([token.text for token in doc])
print([token.pos_ for token in doc])
# ['DET', 'VERB', 'DET', 'NOUN', 'PUNCT']
# Fine-grained part-of-speech tags
print([token.tag_ for token in doc])
```

```
['the', 'pilot', "'s", 'approach', 'was', 'off', '-', 'center', '.', 'He', 'still', 'manage',
'd', 'a', 'three', '-', 'point', 'landing']
['DET', 'NOUN', 'PART', 'NOUN', 'AUX', 'ADV', 'PUNCT', 'NOUN', 'PUNCT', 'PRON', 'ADV', 'VER',
'B', 'DET', 'NUM', 'PUNCT', 'NOUN', 'NOUN']
['DT', 'NN', 'POS', 'NN', 'VBD', 'RB', 'HYPH', 'NN', '.', 'PRP', 'RB', 'VBD', 'DT', 'CD', 'HY',
'PH', 'NN', 'NN']
```

NOUN PHRASE CHUNKING USING SPACY

```
In [5]: doc = nlp("Neural Network model predictions get better and better")
        for np in doc.noun_chunks:
            print(np.text)
```

Neural Network model predictions

```
In [6]: doc = nlp("the pilot's approach was off-center. He still managed a three-point landing")
        for np in doc.noun_chunks:
            print(np.text)
```

the pilot's approach
He
a three-point landing

```
In [7]: doc = nlp("Autonomous cars shift insurance liability toward manufacturers")
        for np in doc.noun_chunks:
            print(np.text)
```

Autonomous cars
insurance liability
manufacturers